STAMP: Selective Task-Aware Mechanism for Text Privacy

1. Introduction

LLMs are often used on text that contains private details such as names, places, dates, and ID numbers. If this text goes to a remote model, the server sees everything. So, we need protection on the user's side, before the text leaves the device.

The paper studies two uses of this idea:

- **Inference-time context privacy:** the text is privatized right before it is sent to a remote LLM.
- **Privacy-preserving rewriting:** the text is changed locally before it is stored or shared.

The paper uses local differential privacy (LDP). Noise is added on the user's device, so the server only sees the privatized text $\tilde{c}$. Informally, the server should not be able to tell reliably whether a specific token was in the original text.

Problems with earlier methods

1. **Randomized response:** replaces tokens with random tokens. The text becomes unnatural and the utility drops.
2. **Laplace or Gaussian noise on embeddings:** embeddings are not equally sensitive in all directions, so equal noise in all directions is a poor fit.
3. **Direction-only noise used uniformly:** keeps the magnitude, but if the same budget is used for every token and the decoding rule does not match, meaning is still damaged.
4. **Uniform budget for all tokens:** harmless words and important words get the same noise.
5. **Heuristic selective methods** (frequency, part of speech, information content): these ignore the **task**.

So, the authors identify a basic trade-off:

Strong noise → better privacy, but the LLM cannot answer Weak noise → better answers, but private tokens leak

The main idea of the paper is to spend the privacy budget selectively. Protect the tokens that are private and not needed, and leave alone the tokens that are needed and not private.

Why the task matters (the paper's examples)

- "Einstein" is essential for "Who developed the theory of relativity?" but irrelevant for "When was the Nobel Prize established?"
- In customer support, "Alice Johnson" or "ID: 123-45-6789" is very sensitive but not needed, so it should get strong noise.

The three contributions

1. **Selective, task-aware privacy allocation.** Tokens are split by task importance and privacy sensitivity, and each group gets its own budget.
2. **Geometry-aligned perturbation (Polar mechanism).** Only the direction of the embedding is perturbed, and decoding uses cosine nearest neighbor.
3. **Empirical evaluation** on SQuAD, Yelp, and AG News at matched per-token budgets.

Figure 1 (overview)

Tokens → Grouping (G1–G4) → Budget allocation (ε1–ε4) → Polar mechanism → Private tokens

Figure 1(b) is the 2×2 table:

	Task importance: High	Task importance: Low
Privacy importance: High	G1: Low ε	G2: Lowest ε
Privacy importance: Low	G3: Highest ε	G4: High ε

Remember: lower ε means more noise.

Figure 2 (fantasy example)

Task: “How does the children’s laughter feel in the heavy air?”

Setting	Text	Answer
Baseline (no privacy)	Original text	Fragile ✓
Uniform Laplace (ε = 450)	Almost fully garbled	Aggressive ✗
STAMP + Polar (ε ≈ 450)	Names changed, story words readable	Fragile ✓

This is one chosen example that shows the idea. The real evidence is in Section 5.

2. Preliminaries and Related Work

Section 2 sets up the problem and the privacy definitions, then reviews earlier work.

2.1 Preliminaries on LDP for text

Setting.

A user has a context $c = (w_1, \ldots, w_n)$ with tokens from a vocabulary V. A local mechanism M produces a privatized version $\tilde{c}$. The server sees only $\tilde{c}$.

Embedding.

Each token has an embedding $e(w) \in \mathbb{R}^d$, and its direction is $\hat{e}(w) = e(w) / \|e(w)\|_2$.

Token-level mechanism.

$M : V \rightarrow Y$ *is applied independently to each token*:

$M^{(n)}(c) = (M(w_1), \ldots, M(w_n))$

Definition 1: (ε, δ)-LDP (token level).

For every pair of tokens w, w′ and every output set S:

$$Pr[M(w) \in S] \leq e^{\varepsilon} \cdot Pr[M(w') \in S] + \delta$$

Smaller ε (and small δ) means stronger privacy.

Definition 2: (ε, δ)-metric LDP.

The bound becomes $e^{\varepsilon \cdot d(w,w')}$, where d is a distance between tokens. Similar tokens (small d) must look alike, and far tokens may differ.

Why this fits text: "doctor" and "physician" should be harder to tell apart than "doctor" and "banana." The distance d can be cosine, Euclidean, or from contextual representations.

2.2 Related work on text privacy

The paper groups earlier work in four points.

1. **Words differ in privacy risk.** Names, emails, and locations reveal a lot. Stop words reveal little. Many methods ignore this.
2. **Isotropic embedding perturbation.** Gaussian or Laplace noise assumes all directions are equal, but embedding spaces are anisotropic (some directions carry fine meaning, others carry frequency or style).
3. **Directional privatization.** There are vMF-based and spherical-cap methods with DP guarantees (Weggenmann and Kerschbaum, 2021). Word-level metric DP often adds noise and then decodes by cosine nearest neighbor. But uniform direction noise that ignores the task, with a mismatched decoding geometry, still disrupts fine relations.
4. **Uniform budget allocation.** Most work gives every token the same budget. The closest earlier work (Meisenbacher et al., 2024) redistributes the budget by intrinsic text scores, but it is still **task-agnostic**.

What STAMP changes:

- It models both sensitivity and task relevance.
- It treats privacy as a contextual choice about what to hide and what to keep.
- It separates the allocation strategy from the perturbation mechanism. Polar is used here, but other mechanisms with a tunable strength parameter (Carvalho et al., 2023; Feyisetan et al., 2020) could be plugged in.

3. The STAMP Framework

Section 3 explains the allocation side: how tokens are grouped and how privacy budgets are assigned.

3.1 Overview

STAMP splits tokens along two binary axes:

- **Explicit sensitivity** (PII or NER labels)
- **Task-dependent importance** (similarity to a task or query representation)

Two principles govern the budgets:

1. Sensitive tokens get the strongest protection.
2. Among non-sensitive tokens, more important ones get the least noise.

Each token w is mapped to a group $c = g_T(w) \in \{1, 2, 3, 4\}$ by a public, task-specific map g_T. Budgets are set per group.

Formal guarantees

Definition 3: Task-aware metric LDP.

For a task T with token groups V_1^T, …, V_4^T and budget vector $\varepsilon^T = (\varepsilon^{(1)}, \ldots, \varepsilon^{(4)})$: for every group c, all tokens w, w′ in group c, and every output set S,

$$Pr[M(w) \in S] \leq e^{\varepsilon^{(c)}} \cdot d_u(w, w') \cdot Pr[M(w') \in S]$$

Theorem 1 (privacy guarantee). If T and g_T are public, STAMP satisfies task-aware metric LDP with budget vector ε^T. Each group is $(\varepsilon^{(c)}, 0)$ −metric LDP. If ε = 0 for a token, nothing about it is revealed (it is effectively masked).

Theorem 2 (context level). Tokens are privatized independently, so the guarantees add up over a sequence:

$$Pr[M^{(n)}(w) \in S] \leq exp(\Sigma_i \varepsilon^{(ci)} \cdot d_u(w_i, w'_i)) \cdot Pr[M^{(n)}(w') \in S]$$

3.2 Group map and budget allocation

Designing the group map.

- Sensitivity: found with named-entity or PII recognizers.
- Task relevance: from gradient-based saliency, attention scores, or cosine similarity to a task or query embedding.
- g_T is public, so the grouping policy itself leaks nothing and the guarantees compose cleanly.

Allocating budgets. The rule of thumb is that the budget should increase with task importance and decrease with sensitivity. The experiments use:

$$\varepsilon^{(1)} : \varepsilon^{(2)} : \varepsilon^{(3)} : \varepsilon^{(4)} = 2 : 1 : 4 : 3$$

Group	Meaning	Budget
G1	Sensitive and task-critical	Moderate (balance)
G2	Sensitive but less important	Smallest (most noise)
G3	Low sensitivity, high importance	Largest (least noise)
G4	Low sensitivity, moderate importance	High

The paper says this ratio is only a transparent example of the ordering. It is not claimed to be optimal.

4. Polar Mechanism under Metric LDP

Section 4 explains how one token embedding is noised. The idea is to **match the noise geometry to the decoding geometry**.

4.1 Splitting the embedding

e = r · u, r = ‖e‖₂ (magnitude), $u = \frac{e}{\|e\|_2} \in S^{d-1}$ (direction)

- **(a) Magnitude:** Laplace noise, $r' \sim Laplace(r, b_r)$ with $b_r = \frac{\Delta r}{\varepsilon_r}$. This is $(\varepsilon_r, 0)$-metric LDP under $d_r = |r - r'|$.

- **(b) Direction:** von Mises–Fisher noise, $u' \sim vMF(\mu = u, \kappa). \kappa \geq 0$ controls the concentration (bigger κ means less noise).

Theorem 3 (vMF is metric LDP). With density $f(y|u) = C_d(\kappa)\, exp(\kappa\, u^{\mathrm{T}} y)$, the mechanism satisfies

$$\Pr[M(u) \in S] \leq e^{\kappa \|u - u'\|_2} \cdot Pr[M(u') \in S]$$

So it is ε-metric LDP with ε = κ under the chordal distance $\|u - u'\|_2$. Combining (a) and (b) gives a two-budget guarantee $(\varepsilon_r, \varepsilon_u)$.

4.2 Semantic decoding

The noisy vector $e' = r'u'$ may not match any token. So it is decoded to the token with the highest cosine similarity:

$$Decode(e') = argmax\ over\ v \in E\ of\ \frac{e'^{\mathrm{T}} v}{(\|e'\|_2 \|v\|_2)}$$

Radial invariance. Decoding depends only on direction, so changing ‖e′‖ cannot change the result. This motivates dropping the magnitude noise.

4.3 Normalized Polar mechanism

$$\bar{e} = \frac{e}{\|e\|_2}, \quad u' \sim vMF(\bar{e}, \kappa)$$

- It gives (0, ε_u)-metric LDP, so it is a single-budget, direction-only mechanism.
- The magnitude is perfectly private, since the output always has norm 1 and does not depend on r.

4.4 Computational cost

- Exact nearest-neighbor search costs $O(|V| \cdot d)$. Approximate search (HNSW, FAISS) is sublinear in |V|.
- **Proposition 1.** Cosine decoding, geodesic nearest neighbor, and chordal nearest neighbor are the same rule on the unit sphere. So, the decoding geometry matches the privacy metric.

4.5 Algorithm 1 (STAMP–Polar)

For each position i:

1. $c_i \leftarrow g_T(w_i)$ $(find\ the\ group)$
2. $\bar{e}_i \leftarrow \frac{e(w_i)}{\|e(w_i)\|_2}$ $(normalize)$
3. $\kappa_i \leftarrow \varepsilon_u{}^{(ci)}$ $(budget\ of\ the\ group\ is\ the\ concentration)$
4. $Sample\ u'_i \sim vMF(\bar{e}_i, \kappa_i)$
5. $w'_i \leftarrow argmax\ over\ v\ of\ \frac{u'_i{}^T v}{\|v\|_2}$ $(decode)$

Span-level constraint. If a detector finds a multi-token entity (for example "New York"), the whole span goes to the same group. Small boundary errors are tolerated because Polar only changes direction and keeps local semantic neighborhoods.

Figures 3 and 4

- **Figure 3 (pipeline):** Context and Task are tokenized and embedded. Then Privacy and Task-Aware Categorization, then STAMP and Polar budgeting, then the perturbed embedding, then the private context.
- **Figure 4:** shows the unit circle, with the normalized embedding vector rotated by an angle θ. Only the angle changes.

5. Experiments

Section 5 tests whether STAMP and Polar work. It asks two questions:

1. At the same privacy budget, does Polar (vMF) beat isotropic Laplace?
2. At the same privacy budget, does STAMP beat a uniform allocation?

5.1 Experimental setup

Dataset	Task	Metric
SQuAD	Question answering	Cosine similarity of answers (Sentence-BERT); EM also mentioned
Yelp	Sentiment rating	Accuracy
AG News	Topic classification	Accuracy

- **QA evaluator:** GPT-4 (main text). Appendix C says gpt-4o-mini was used for answering. Check which one is correct before citing.
- **Classification:** existing models trained on the original (non-private) training sets. Privatization is applied only at test time (inference).
- Results are averages over **three runs**. No rewriting unless stated.
- **Grouping:** $importance = cos(token, query) \geq \tau$, with $\tau = 0.5$. Sensitivity = NER or PII tags for Person, Location, Organization, and Numeric Identifier.
- **Matched privacy budget:** both mechanisms satisfy ε-metric LDP for the same ε, so a utility gap is due to geometry or allocation, not extra budget.

Choosing τ (Figure 5). The authors tried τ from 0.3 to 0.7 and measured the cosine similarity between the original and privatized contexts. It flattens around τ = 0.5, so 0.5 became the default. Note that this measures similarity of contexts, not task accuracy.

5.2 Polar vs Laplace (Table 1)

Both use d = 768 and cosine decoding. Rows are group budgets {G1, G2, G3, G4}.

Budgets {G1, G2, G3, G4}	SQuAD Polar	SQuAD Laplace	Yelp Polar	Yelp Laplace	AG Polar	AG Laplace
{150, 50, 450, 350}	0.393	0.325	0.360	0.220	0.540	0.560
{200, 100, 500, 400}	0.470	0.334	0.380	0.140	0.640	0.580
{250, 150, 550, 450}	0.587	0.335	0.480	0.180	0.680	0.560
{300, 200, 600, 500}	0.654	0.341	0.560	0.200	0.760	0.520
{350, 250, 650, 550}	0.833	0.343	0.560	0.140	0.800	0.520
Non-private baseline	0.839		0.580		0.920	

Main result: Polar improves smoothly as the budget grows and approaches the non-private baseline. Laplace stays flat and low.

Reading notes:

- On SQuAD at the top budget, Polar (0.833) is almost at the baseline (0.839).
- On AG News in the first row, Laplace (0.560) is slightly higher than Polar (0.540). So the "Polar always wins" claim is not true for every cell.
- Yelp has 5 rating classes, so chance is about 0.20. Laplace is at or below chance in several rows. That matches the paper's claim that Laplace collapses at low budgets.
- Laplace on AG News (0.52 to 0.58) is above the 4-class chance level of 0.25, so "collapse to chance" does not describe AG News.
- The budget rows increase by a fixed +50 in every group. They do not follow the 2 : 1 : 4 : 3 ratio in Section 3.2 (the first row is 3 : 1 : 9 : 7).

5.3 STAMP vs Uniform (Figure 6)

At matched per-token budgets, STAMP beats a Uniform scheme, especially in the **low-to-mid privacy regime**.

- STAMP keeps important content and puts the noise on spans that are sensitive but task-unimportant.
- Utility is highest when only a small subset is perturbed. As more groups are protected (one, two, three, then all four), performance drops steadily. Uniform (the same budget for every token) is the lower envelope. It improves only as ε grows and the noise disappears.
- Question-conditioned importance (SQuAD) increases STAMP's gain. Fixed prompts (Yelp, AG News) reduce it, because the group sizes stay almost constant.
- STAMP gives a controllable knob over which content to protect.

The Figure 6 variants (private-only, private-low, heuristic) are different ways to choose which groups receive noise. The figure shows accuracy against the base budget ε_0 on all three tasks.

5.4 Computational overhead

Step	Cost
Grouping and budgeting	O(n) in tokens. About 0.002 s per SQuAD example (about 180 tokens).
vMF sampling	O(d) with efficient rejection sampling, like Gaussian or Laplace.
Runtime per example	STAMP–Polar 35.16 s (195 ms/token) vs Laplace 34.54 s (192 ms/token)

Main result: the privacy step adds almost no extra time compared with uniform isotropic baselines. Most of the time is spent on the model.

6. Conclusion

STAMP is a task-aware mechanism for text privatization under LDP. It combines geometry-aware perturbation (Polar) with task-dependent budget allocation. It outperforms Uniform at matched per-token budgets, and Polar beats isotropic Laplace. Future work covers dynamic tasks and sequence-level dependencies.

7. Limitations

- Needs a fixed task representation at inference time. This may not hold for interactive, open-ended, or multi-turn use.
- Static embedding similarity may miss deeper importance (syntax, discourse, outside knowledge).
- $d = 768$ needs large per-token ε to keep utility, which is a known issue in text LDP. The contribution is relative: under a fixed budget, protection is moved to where it is needed.
- Depends on the quality of the task and sensitivity oracle (PII boundary errors).
- No handling of long-range dependencies between tokens in the budget, so over- or under-noising can happen.

8. Ethical Considerations

- STAMP can allow use of sensitive text (email, finance) without exposing the raw content.
- Risks: uneven protection across demographic or domain categories, weak guarantees if budgets are set carelessly (false sense of safety), and no handling of fairness or downstream harms.
- It is a methodological tool. Responsible use needs auditing, fairness checks, and clear reporting of privacy parameters.

Appendix A: Threat Model

Setting. Each user holds a private text C and applies M locally to get $\tilde{C} = M(C, T)$. Only $\tilde{C}$ is shared.

Adversary. An honest-but-curious server. It follows the protocol but tries to infer information from $\tilde{C}$.

Adversary goals:

1. Identify specific sensitive tokens or spans (names, locations, medical terms).
2. Infer user attributes correlated with C (demographics, preferences).
3. Fully or partly reconstruct C.

Adversary knowledge: the mechanism and its parameters, the vocabulary and embeddings, the public map g_T , and any background knowledge. It sees only the privatized output.

Assumptions: randomization is local and independent per position. The RNG is not controlled by the adversary. g_T and the hyperparameters (κ or ε per group) are **public**, so privacy does not depend on keeping them secret.

Out of scope: side channels (timing, memory), compromised clients, a corrupted RNG, and leakage after aggregation that is not linked to the local mechanism.

Guarantees in scope: the metric LDP guarantee of the angular channel (Theorem 3) and its composition over a sequence (Theorem 2). Experiments report the average per-token budget $\bar{\varepsilon}$ and group counts (#G1 to #G4).

Appendix B: Proofs

Result		Proof idea
Theorem (B.1)	**1**	Fix a group c. The map g_T is public and deterministic, so restricting to that group keeps the worst-case guarantee. Apply Theorem 3 with κ_c .
Theorem (B.2)	**2**	Per-token channels are independent, so probabilities multiply and the exponents add: $\Sigma\ \varepsilon_t\ d(x_t, x_t')$.

Result	Proof idea
Theorem 3 (B.3)	$log[\frac{f(y\|\mu)}{f(y\|\nu)}] = \kappa(\mu-\nu)^{\mathrm{T}}y \leq \kappa\|\mu-\nu\|_2$ by Cauchy–Schwarz and ‖y‖ = 1. The constant $C_d(\kappa)$ cancels. Since $d_2 = 2\,sin\left(\frac{\theta}{2}\right) \leq \theta = d_g$, the bound also holds for geodesic distance.
Proposition 1 (B.4)	For fixed κ, $log\, f(y\|\mu(w)) = \kappa\,\mu(w)^{\mathrm{T}}y + constant$, so maximizing the likelihood is the same as maximizing cosine.

Appendix C: Additional Experimental Results

C.1 Prompts

Coherence repair (optional rewrite step). gpt-4o-mini, temperature 0.2. The system prompt says: act as a careful editor, rewrite into coherent, grammatical English, keep the meaning and tone, do not add external facts, do not invent names, dates, locations, or entities, do not expand abbreviations, and keep unusual tokens as they are. Return only the rewritten passage.

Answer generation (evaluator). gpt-4o-mini, temperature 0.2, with the instruction to answer from the given context in one word.

C.2 Fantasy SQuAD dataset

Problem: LLMs may have seen SQuAD in pretraining, so they can answer from memory even if the context is destroyed. This confounds the test.

Solution: the authors had GPT-4 generate a synthetic dataset of fictional passages (invented cities, events, and species) with 5 questions each. Every proper noun and fact is made up, so any correct answer must come from the privatized context.

Effect: this gives a rigorous lower-bound estimate of utility preservation.

C.3 Qualitative examples (Figures 7 to 10)

All examples are shown at a matched average budget ($\bar{\varepsilon}$ about 540 for the Yelp and Business examples, about 490 for the Sci/Tech example), after a coherence-repair rewrite.

Figure	Task	STAMP and Polar keeps	Uniform and Laplace
7	Yelp rating	Words like “underwhelmed” and the rating cue “three stars”	Unintelligible
8	AG News (Sci/Tech)	“Peptides”, “chemistry researcher”, “protein”	Unrelated tokens
9	Yelp rating	“video games”, “ticket projects”, “packed” (positive sentiment)	Nonsense tokens
10	AG News (Business)	“law strengthened job protections”, “active duty”, “(USERRA)”	Complete obfuscation

These are selected examples, so they illustrate the effect but do not prove it. The numbers in Table 1 and Figure 6 are the main evidence.